

# Bayesian Approaches for Imputing Missing Data in Environmental Studies: Hierarchical Models and Kalman Filter Techniques

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## Abstract

Missing data present significant challenges in environmental sciences, particularly when accurately characterizing environmental dynamics. Imputing missing values in nonlinear systems, common in environmental measurements, requires incorporating all relevant and available data. Without a comprehensive approach, imputed measurements risk failing to capture the true environmental dynamics or introducing biases into subsequent analyses. In this paper, we present three Bayesian approaches to impute missing data, explicitly incorporating measurement uncertainties and uncertainties arising from model assumptions while utilizing all available data. First, we demonstrate Bayesian hierarchical regression, which effectively accounts for the nested structure typical of environmental datasets. Next, we explore hierarchical regression models incorporating change points to represent gradients or boundaries within the data. Finally, we examine Bayesian implementations of Kalman filters enhanced by Bayesian variable selection methods.

*Keywords:* Bayesian models, hierarchical models, missing data, environmental science, uncertainty quantification, imputation

## Introduction

Measurement data often contains missing values, and imputation becomes particularly complex for data sampled in nonlinear mediums or gradients (see d'Auria *et al.*, this volume) that defy easy mathematical characterization, such as those encountered in environmental studies.

Statistical approaches to imputation should leverage all relevant data, but this inevitably increases model dimensionality and complexity, making it essential to quantify uncertainty accurately. Since statistics is the art of quantifying uncertainty, the Bayesian framework is particularly well suited to this task. Bayesian methods that are typically implemented via hierarchical (multilevel) models incorporate assumptions and prior information into probability distributions to offer a coherent, nuanced representation of complexity and uncertainty.

Bayesian statistics, notable for its ability to update event probabilities based on prior information and observed data, offers a versatile framework for data analysis. Its incorporation of prior information distinguishes it from frequentist statistics, facilitating the integration of outcomes from previous studies or expectations into analyses. Particularly advantageous for relatively small datasets, such as those typical in environmental studies, the Bayesian approach allows for dynamic updating of results with new data, enhancing adaptability and robustness.

The unified framework of Bayesian statistics addresses a wide range of statistical challenges, such as parameter estimation, model comparison, and hypothesis testing. It excels in handling complex models, including hierarchical structures and missing data problems, while offering a natural approach to quantifying uncertainties through probability distributions. Additionally, the ability to define all components of a model within the same framework provides a holistic approach, facilitating comprehensive assessment of uncertainties associated with both

measurement datasets and the model itself. Readers unfamiliar with Bayesian approaches and hierarchical models can refer to Kruschke (2015), Gelman *et al.* (2013) or Gelman and Hill (2006) for further insights.

In this context, we briefly present three examples from our previous works that apply Bayesian techniques to impute missing data. These methods assume that missing values depend on available information and treat each observation as a random sample contributing to the overall result. First, we describe two multilevel Bayesian approaches, which naturally arise when modeling missing data as conditional on observed covariates. A multilevel model specifies hierarchical distributions at both upper and lower levels that inform one another, integrating information across strata and improving predictive power by shrinking extreme low-level estimates toward the group mean (Kruschke and Liddell, 2018). Finally, we showcase a Bayesian Kalman-filter solution for predicting missing values.

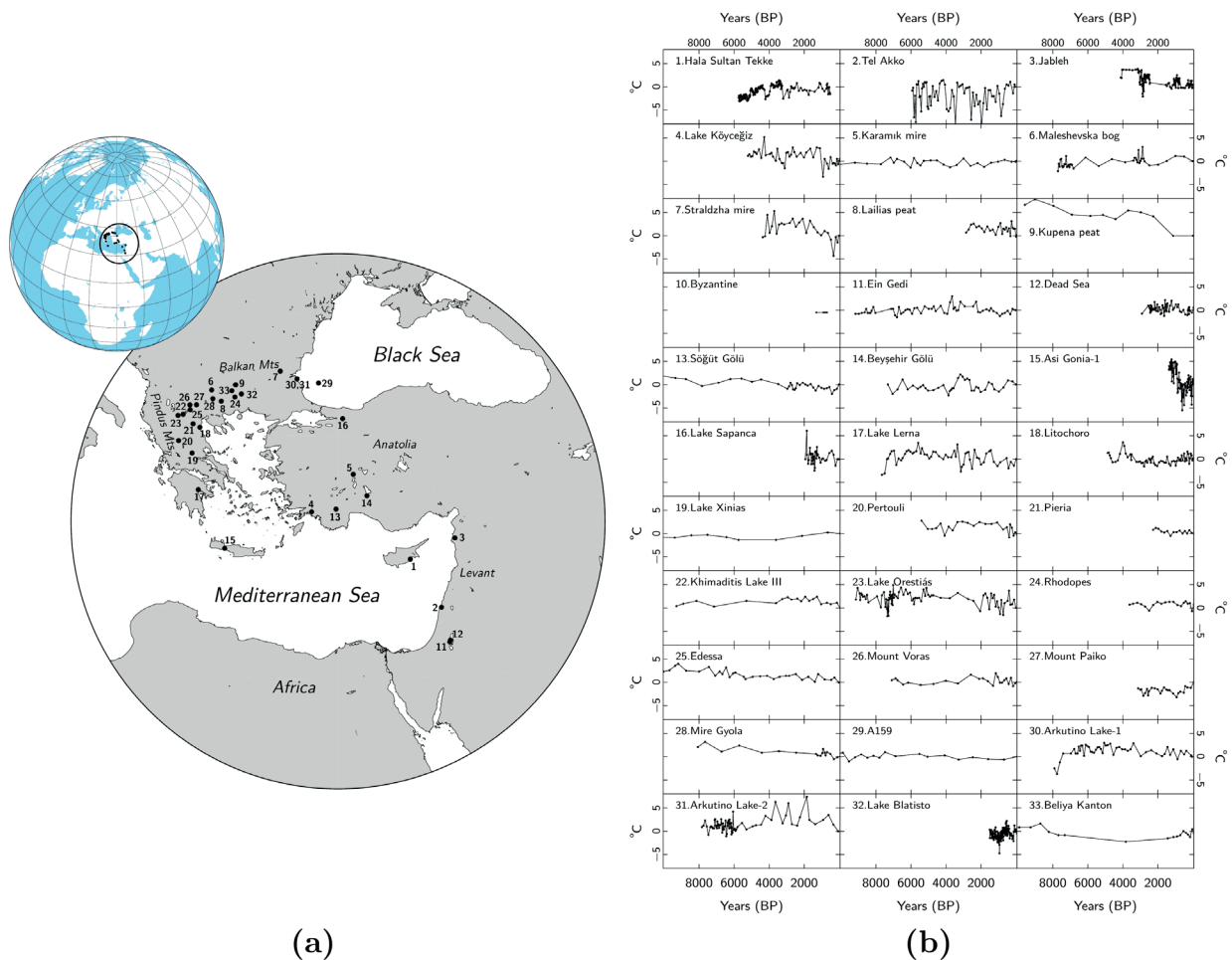
## Examples of Bayesian Models

### Hierarchical Models

#### *Bayesian Hierarchical Regression*

Consider a dataset of direct or proxy measurements of the same underlying quantity, collected from multiple locations over time, or depths, or across gradients of another variable. For the purpose of this study, let's assume that these measurements are expected to exhibit similar patterns, potentially due to collection in comparable environments or clustering within regions anticipated to display analogous trends. It is important to note that some of these measurements may be more complete than others, characterized by higher resolution in time or depth, or simply possessing more data points. For example, different temperature reconstructions of the Holocene in the eastern Mediterranean from pollen counts depicted in Figure 1 (Ön *et al.*, 2023) exemplify such variations in completeness.

In our study, our objective was to reconstruct an “average” temperature anomaly in the eastern Mediterranean region over the past 10,000 years through a meta-analysis, acknowledging the different resolutions, variabilities, and trends observed in each dataset when viewed collectively (Ön *et al.*, 2023). We utilized temperature reconstructions derived from pollen counts from various sites (Figure 1). To align with our research objectives, we segmented the data into 300-year intervals. Subsequently, we applied a Bayesian hierarchical linear regression model to analyze the dataset. In this hierarchical regression model, regressions are calculated for all individual samples (Figure 2a), while also incorporating an upper-level regression within the same model (Figure 2b).

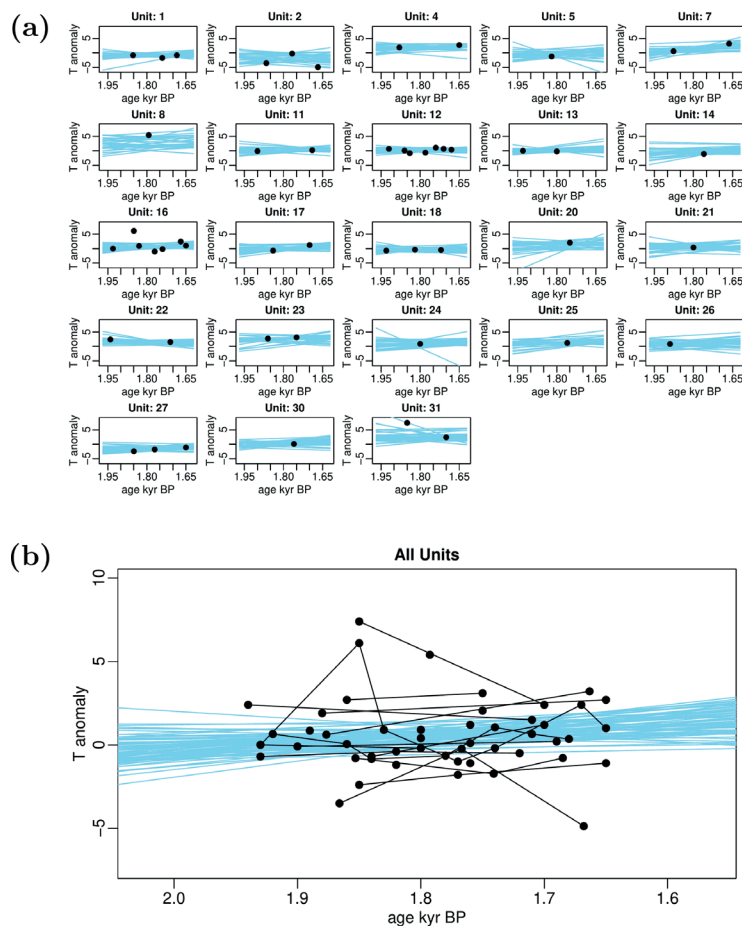


**Figure 1.** (a) Locations of pollen sites in the eastern Mediterranean. (b) Various temperature reconstructions derived from pollen counts. For details and references regarding each dataset, please consult Ön *et al.* (2023), from which the figure is adapted.

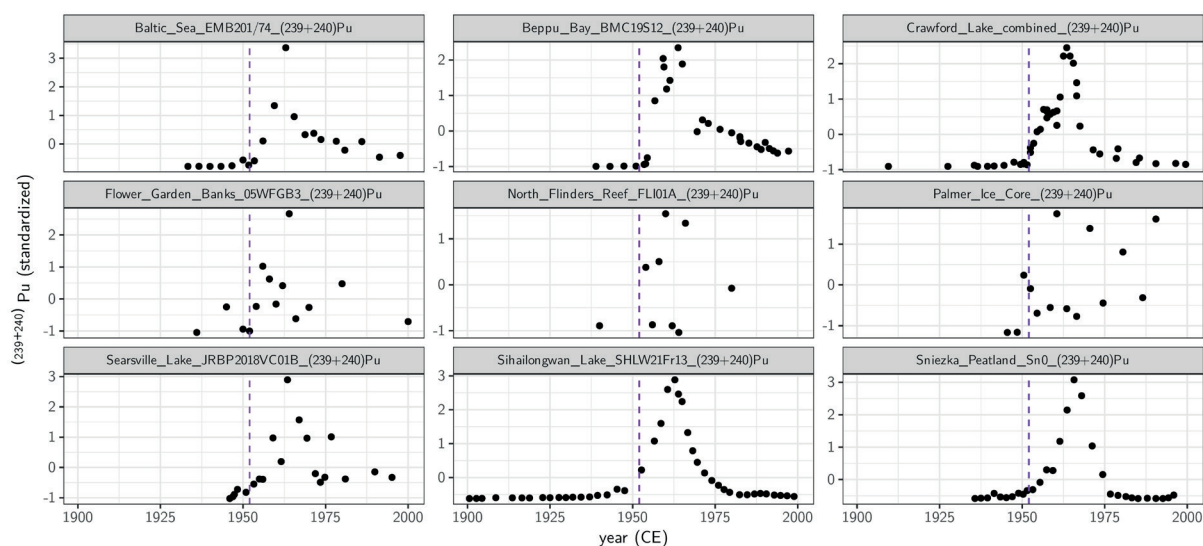
A key advantage of this multilevel model is partial pooling, where information flows bidirectionally between individual-level and population-level parameters. For example, units with sparse data (*e.g.*, Unit 5 and Unit 20 in Figure 2a) borrow strength from the overall distribution, yielding reasonable regression fits and well-calibrated uncertainty estimates. Units with outlying trends (*e.g.*, Unit 31) can still deviate when their data provide strong evidence, since the hierarchical prior shrinks—but does not override—their estimates. Intermediate cases (*e.g.*, Unit 16) strike a balance between local variation and the population mean, avoiding overfitting while preserving genuine signal. In a setup like this (see Figure 2), the hierarchical model naturally regularizes noisy or under-sampled datasets through shared regression parameters and can, if necessary, impute missing observations seamlessly within the same inference framework.

### *Bayesian Hierarchical Change Point Analysis*

Consider again multiple measurements collected within a localized region or across similar environments exhibiting comparable patterns over time, depth, or other dimensions. These patterns manifest as shifts in the mathematical functions describing the data. For example, refer to the  $^{(239+240)}\text{Pu}$  measurements presented in Figure 3, obtained from various environments (for dataset references, see Waters *et al.*, 2023 and references therein).



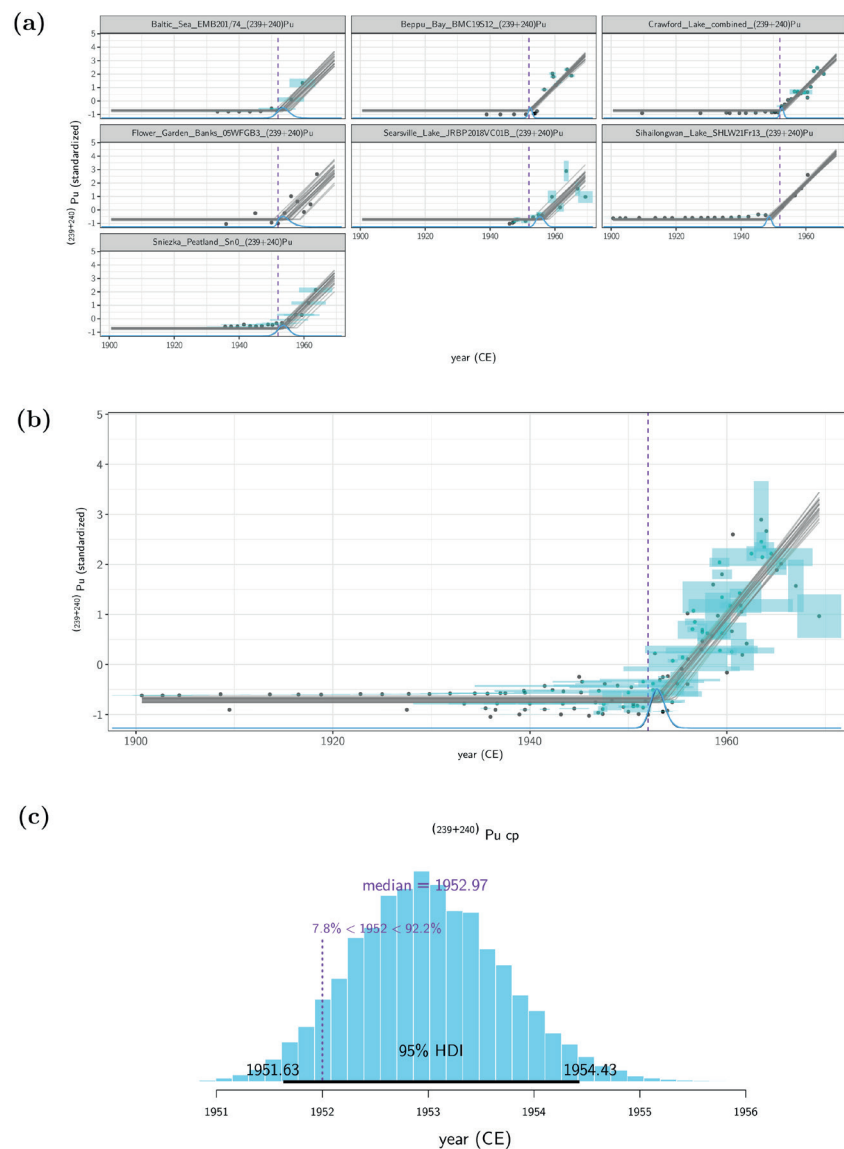
**Figure 2.** Linear regression curves of the hierarchical model for the period between 1650–1950 calibrated years before present. (a) For low-level regression parameters, (b) For high-level regression parameters. Each plot in (a) represents a different measurement in the region depicted in Figure 1a. The blue bunch of lines represents random regression solutions generated from posterior draws. The figure is adapted from Ön *et al.* (2023).



**Figure 3.**  $(239+240)\text{Pu}$  measurements from various environments and locations serve to pinpoint the onset of the Anthropocene. The purple dashed line denotes the year 1952 CE, marking the occurrence of the first H-bomb testing. The figure is adapted from Ön *et al.* (2025).

Our objective with these data was to critically evaluate the designation of 1952 CE as the onset of a potential Anthropocene epoch using a Bayesian hierarchical change point model.

As anticipated, the measurements are imperfect due to several factors, including sample quality and resolution, uncertainties inherent in the measurement processes, and variations in the preservation of proxy signals under differing environmental conditions. While some records clearly show an abrupt increase followed by a somewhat gradual decline, others exhibit less distinct patterns.



**Figure 4.** The hierarchical change point analysis results for  $^{(239+240)}\text{Pu}$  data are presented across distinct datasets (a), for the overall change point (b), and the posterior distribution of the overall change point (c). Our assumption for this dataset was a constant level followed by a ramp. Each dataset was truncated after approximately 1965 CE, aligning with the model described in Öñ *et al.* (2025). Cyan boxes in (a) and (b) denote one standard deviation uncertainty intervals associated with dates and measurements. Gray lines represent random solutions generated from posterior draws. The purple dashed lines indicate 1952 CE, while the black bar below the posterior distribution in (c) denotes the 95% highest density interval. The figure is adapted from Öñ *et al.* (2025).

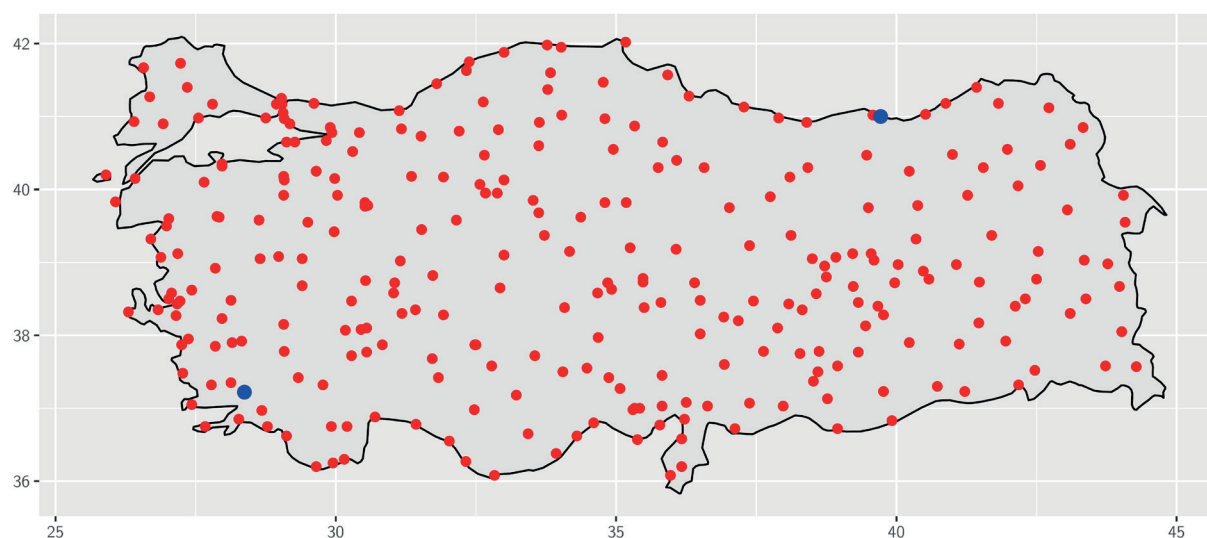
To delineate the start of the increase, we removed the data after around 1970 CE (for details, consult Öñ *et al.*, 2025). Applying a hierarchical change point analysis using an errors-in-

variables approach, which accounts for the associated uncertainties in both proxy and dating measurements, can effectively regularize the data (Figure 4). Consequently, it can serve as a useful tool for imputing missing data in this type of dataset.

Notice how uncertainties are integrated into the model and how their impact is reflected in the increasing uncertainty of the regression and change-point parameters for more variable records (see Figure 4). Furthermore, even when the exact location of the change point is unclear in certain datasets, the hierarchical model helps pinpoint its location by shrinking the posterior distributions toward the group mode through shared information between higher-level and lower-level parameters.

### Kalman Filter Through Variable Selection

Missing data in environmental time series present a significant challenge, particularly for time series analyses, which often require evenly spaced data. For example, missing data may arise in certain gauge or climate stations, while neighboring stations possess complete datasets. Imputation methods based solely on the inherent structure of the time series with missing data may overlook the non-stationary nature of environmental data. Therefore, filling these gaps through a unified approach that considers both the inherent structure of the data with missing values and the available complete data should be the preferred method.



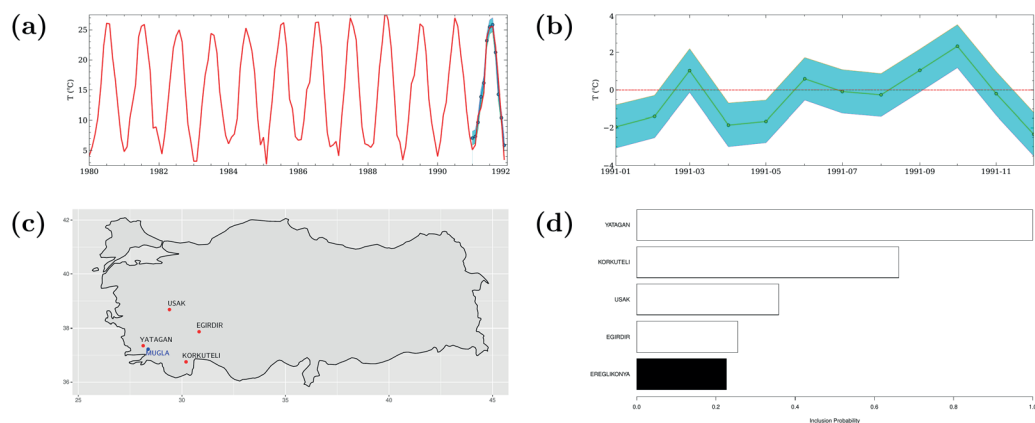
**Figure 5.** The locations of the meteorology stations used in the study. Muğla and Trabzon stations are given in blue color.

Turkey can be described as a high plateau peninsula, with a mean altitude of approximately 1,132 m, covering a relatively large land area of approximately 785,000 km<sup>2</sup>. Primarily, tectonically induced topographic gradients, coupled with proximity to water bodies of varying characteristics, result in diverse climatic conditions throughout the country (Ünal *et al.*, 2003; Erinç, 1984).

In our study, we utilized monthly precipitation and temperature station data from Turkey, sourced from Göktürk (2005)<sup>1</sup>. These meteorological stations offer comprehensive spatial

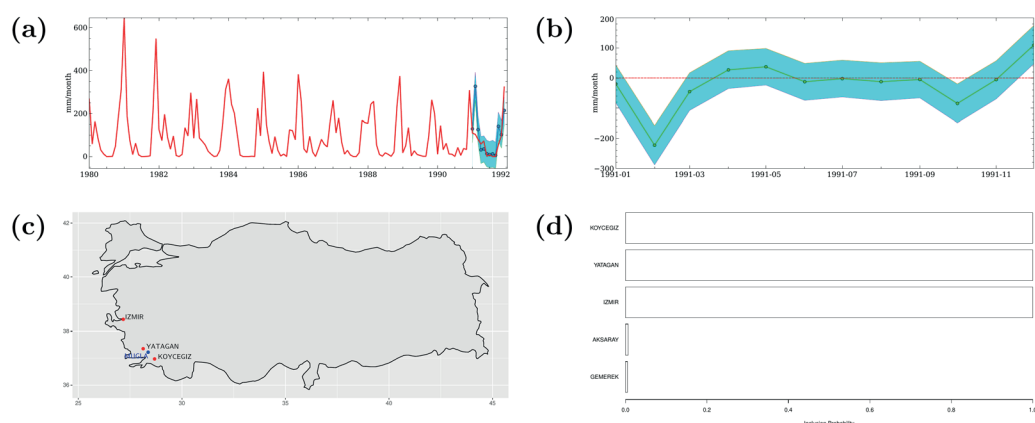
<sup>1</sup> The data may have been derived from daily records, with missing values potentially imputed using expectation maximization, a widely employed technique in climatology (Schneider, 2001). However, this preprocessing step does not affect the objectives of our study. Please note that we cannot definitively confirm the specific imputation method used; this footnote serves as an acknowledgment of uncertainty regarding the imputation procedure.

coverage of the entire country (Figure 5). Specifically, we selected two stations, Muğla and Trabzon, as case studies. Muğla, located in Turkey's southwestern corner and surrounded by high mountains, predominantly receives precipitation from Mediterranean cyclones during winter and experiences a typical Mediterranean climate (wet, mild winters and dry, hot summers). Trabzon, situated on the northern slopes of the east-west trending Eastern Pontides, experiences a Black Sea climate characterized by maritime influences, relatively warm winters, and consistent precipitation year-round.



**Figure 6.** Results of temperature measurements at Muğla station. (a) Imputed data (blue curve) overlaid on measurement data (red curve), with 95% credible interval (cyan shading); (b) Difference between original and imputed data; (c) Map showing Muğla station (blue dot) and predictors with highest inclusion probabilities (red points); (d) Posterior inclusion probabilities of predictors assigned by Markov chain Monte Carlo in the regression model. White bars represent positive coefficients, and black bars represent negative coefficients.

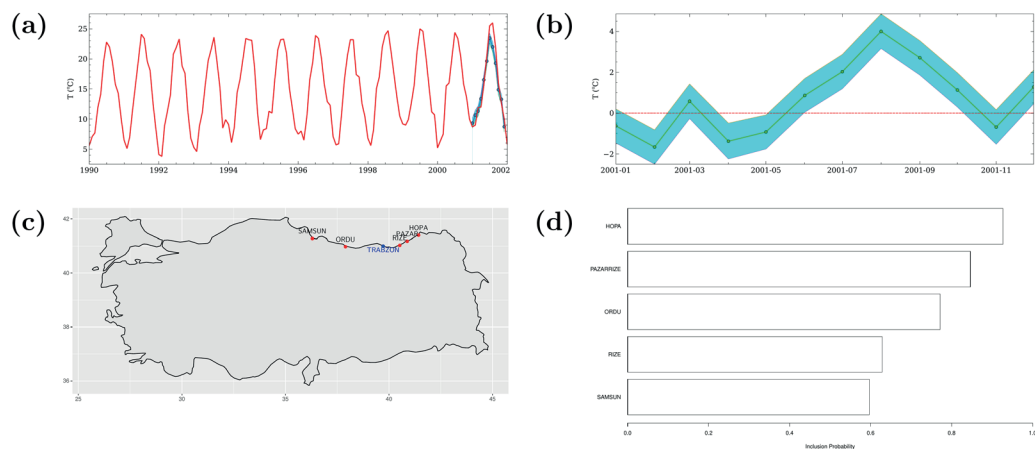
For data imputation, we implemented a time series model along with a regression model using covariates derived from the remaining contemporaneous station data. The time series model incorporates a seasonal process to account for yearly oscillations and an autoregression process of order three.



**Figure 7.** Results for precipitation measurements from Muğla station, similar to Figure 6.

Additionally, we applied the spike-and-slab variable selection (SSVS) technique (Scott and Varian, 2014; George and McCulloch, 1997; O'Hara and Sillanpää, 2009), which serves to reduce the size of the covariate set and effectively cluster covariates showing similar trends

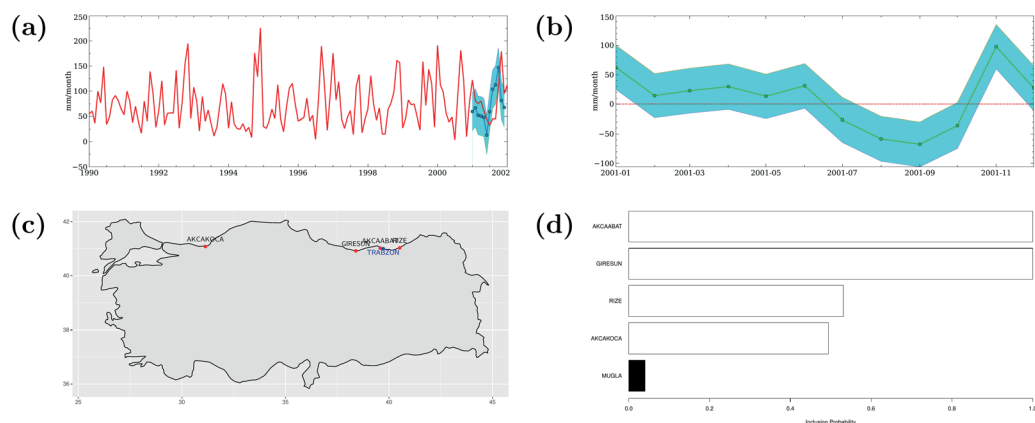
with the target series. In this analysis, the solution is handled through a fully Bayesian model, and therefore the components of the model and weights of each covariate are found through the Bayesian model itself.



**Figure 8.** Results for temperature measurements from Trabzon station, similar to Figure 6.

For imputation validation, we deliberately excluded the entire year 1991 CE from the Muğla monthly temperature and precipitation records. Specifically, for the model to adapt the coefficients, we utilized Muğla data from 1980 to 1990, while employing data from 1980 to 1991 for the remaining stations. The results of the imputation for the entire year of 1991 are depicted for temperature in Figure 6a and for precipitation in Figure 7a.

Similarly, we excluded the entire year 2001 CE from the Trabzon monthly temperature and precipitation data for imputation purposes. Specifically, for the model to adapt the coefficients, we utilized Trabzon data from 1990 to 2000, while employing data from 1990 to 2001 for the remaining stations. The results of the imputation for the entire year of 2001 are presented for temperature in Figure 8a and for precipitation in Figure 9a.



**Figure 9.** Results for precipitation measurements from Trabzon station, similar to Figure 6.

The results of the yearly imputations produced noteworthy results (Figures 6b, 7b, 8b, and 9b). Furthermore, the outcomes of the SSVS method provided insightful findings regarding covariate clustering. The primary covariates identified by SSVS for the regression of Muğla temperature and precipitation (see Figures 6c, 6d, 7c, and 7d) align closely with climatological

clusters identified by Ünal *et al.* (2003), specifically corresponding to the Eastern Mediterranean and Aegean climate zones. Similarly, results from the Trabzon analysis (Figures 8c, 8d, 9c, and 9d) identified a covariate cluster confined to the northern slopes of the Pontides, corresponding to the Black Sea climate zone described by Ünal *et al.* (2003).

### **Concluding Remarks**

There are already dedicated software packages for missing data imputation. However, in the case of more complex datasets, such as those encountered in environmental studies, Bayesian techniques are recommended. These techniques effectively incorporate multiple sources of uncertainty and fully leverage all available data.

Finally, the utilization of Bayesian techniques, particularly those that incorporate total evidence through hierarchical structures, represents a pivotal step forward in addressing the persistent challenges posed by missing data in environmental studies. By explicitly integrating uncertainties and facilitating information sharing across datasets, these sophisticated approaches provide exceptional insights into complex environmental systems.

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